



Stuti Singla

Aerospace Engineering

Indian Institute of Technology Bombay

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22B0046

B.Tech

Female

8373937190

Examination	University	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2026	7.54

PROFESSIONAL EXPERIENCE

Software Engineering Intern | Microsoft

(May '25 - Jul '25)

- Built a telemetry-driven monitoring solution for proactive monitoring across **4K+** nodes in Microsoft's GPU datacenters
- Implemented a secure large-scale event ingestion pipeline handling **25K+** asynchronous hardware and security events per minute using **Redfish APIs**, **Fluentd**, and **Kusto**, integrated into Microsoft's Unified Monitoring Platform
- Automated **BMC** event subscriptions via a **Kubernetes**-based **cronjob**, dynamically querying IPs from the **Kusto**
- Secured **Fluentd** logging with HTTPS and TLS, achieving full **network isolation** per Microsoft's internal standards
- Integrated **QoS health signals**, heartbeat events, and **incident alerts**, enabling proactive monitoring and reliability
- Deployed to production clusters supporting over **30,000** GPUs, with **CI/CD**-driven modular rollout enabling scalable expansion across Microsoft's AI infrastructure.

Research Analyst | Stoolix Technologies

(Jun '24 - Jul '24)

- Analyzed **25+** market reports to conduct comprehensive research and identified key trends within the **F&B** sector
- Improved the decision-making accuracy by **15%** by producing **7** detailed reports, providing insights for management
- Boosted customer satisfaction by **20%** through analysis of consumer behavior from a dataset of **10,000+** respondents

INTERNATIONAL EXPOSURE

Spaceport America Cup 2024 | New Mexico, USA | Rocket team, IITB

(Jun '24)

- Earned **2nd** national, **19th** in **10K ft** commercial motor category worldwide and **35th** overall among **150+** teams
- Co-authored a **70-page project technical report** containing a detailed and thorough analysis of the project's work
- Engaged with **6000+** global rocketeers at an international conference hosted by **ESRA** enhancing industry knowledge

KEY PROJECTS

Image Restoration | Course Project | Machine Learning for Remote Sensing

(Jan '24 - May '24)

- Developed **Deblurring** Model, integrating encoder-decoder architecture with skip connections for spatial retention
- Utilized Python libraries such as **os**, **OpenCV**, **SciPy**, and **PIL** to construct a robust pipeline to blur sharp images
- Optimized model performance, achieving a significant improvement in accuracy from **9.46%** to **76%** over **50** epochs

Bird Species Recognition with CNN | Self Project

(Feb '24-Mar '24)

- Analyzed & preprocessed the **CUB** dataset containing **11,700+** bird images, categorized into **200+** distinct classes
- Attempted to develop a custom CNN model, drawing inspiration from the architecture of **VGG-16** for a given task
- Implemented various pre-trained models like **EfficientNet** and **InceptionNet** and verified their relative performance
- Improved the performance of the final model by **50%**, employing methods like **dropout**, and **data augmentation**

POSITIONS OF RESPONSIBILITY

General Secretary, Hostel 15 | Hostel Affairs | IIT Bombay

(Apr '25 - Present)

Elected representative of **1000+** students, Leading policy and infrastructural changes; H15 Representative in **21+-member** HGSC

- Orchestrated a **45+** member, **3-tier** team collaboration with all institutional authorities to enhance Hostel facilities
- Optimally **managing** all requirements, bills and budget of the Hostel with an overall fund of **INR 15+ Million**
- Leading the direction of a **3-day** fest "**Pandora**", organizing **15+** diverse events, receiving a footfall of over **5000+**

TECHNICAL SKILLS

Programming & Scripting	Python (PyTorch, TensorFlow, OpenCV, NumPy, Matplotlib, scikit-learn, PIL), C++, Bash, SQL, KQL, HTML, LaTeX
Tools & Software	MATLAB, SolidWorks, Fusion 360, AutoCAD, Ansys, OpenRocket, Tracker, OpenVSP, Kubernetes, Fluentd, MS-Office

EXTRACURRICULAR ACTIVITIES

Cultural	Gold Medalist under stage play at the Inter IIT Cultural Meet 6.0 held at IIT Kharagpur - Secured 1st place and a prize of INR 25,000 in stage play at Thirdbell-Mood Indigo 2023
Social	- Raised funds & Awareness for cancer sufferers & the elderly people irrespective of caste - Certified Fortis Pro-Social Peer Moderator for aggression management by Fortis healthcare
Misc.	- Visited the Indian Navy Ship Shikra facility as a part of a departmental Industrial visit - Secured 2nd best interjector title at the Dr. Paulos Mar Gregorious Memorial Debate